

Detection and Classification of Various Types of Leukemia Using Image Processing, Transfer Learning and Ensemble Averaging Techniques

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Abstract— Leukemia, often known as blood cancer, is a frequent and deadly illness. The abrupt rise of White Blood Cells in the blood causes a dangerous illness. Leukemia starts in the bone marrow and spreads throughout the body. It is responsible for the development of a huge number of cells that are aberrant. Leukemia can be categorized into four types. They are Acute myeloid leukaemia, Acute lymphoblastic leukaemia, chronic lymphocytic leukaemia, and Chronic myeloid leukemia. The early detection of leukemia plays crucial role. The detecting process will be more accurate if the diagnosis is computer-assisted. This project aims at developing an approach for detecting leukemia using machine learning techniques such as image processing and transfer learning which provides ways to see if it's a case of AML, ALL, CML, or CLL. Leukemic and healthy lymphocytes have markedly different morphological components. The lymphocyte images have been used to extract a variety of characteristics. The leukemia is further classified into its types by using a hybrid model constructed with the ensemble averaging of Inception, Xception convolutional models.

Index Terms— Leukemia, machine learning, Transfer learning, Inception, Xception.

I. INTRODUCTION

Leukemia is a condition that will affect the blood and bone marrow cells of the physical body and is involved with white blood cells (WBC). This disease has the potential to harm the physical body's systems. Acute and chronic leukaemia are the two major categories of leukemia, which describes how strong it progresses. The infected white blood cells does not perform like healthy white blood cells, however they can act like normal WBC in some cases. As a result, leukemia becomes more severe since it can't be distinguished from the healthy white blood cells. Moreover, depending on the proportions of WBC each leukaemia type has two subclasses which are Lymphoid and myeloid. Hematologists must be able to recognize the presence of leukaemia as well as the different forms in order to minimize medical hazards and provide the appropriate treatment. As a result, adopting intelligent diagnostic methods will enhance and speeds up the process of detection of leukaemia using microscopic blood cell images.

A. Problem Statement

In genal, there might be a lot of problems in healthcare field that are associated with the devices which can give less accurate results, so in order to avoid such false results and obtain the correct and accurate results we are building a program or project which can give the accurate predictions which depends on the input provided by the user to the device

and also on the datasets that are available and used to train the machine. we are making the project which can help all the patients who are in got to know the condition of their health i.e., they have leukemia or not. and if the person has leukemia, then the proposed model will predict the type of leukemia he is suffering with. So, to detect the leukemia in early stages of the symptoms this proposed system will help tons to the varied people including children, adults, teenagers and senior citizens.

B. Objective

The primary goal is to use image processing techniques to create a system that can perform early identification of leukaemia based on blood smear images. If leukaemia is detected, the second goal is to forecast the type of leukaemia by using transfer learning.

C. Scope

The scope of the application is to perform the detection and classify to what type of category the leukemia belongs based on the blood smear images. The system provides a GUI which can be used by common people to take the blood smear image as input and predicts the leukemia.

II. LITERATURE SURVEY

In [1] a hybrid classification methodology for image classification of White Blood Cell Leukemia was described. It is based on extracting features from WBC image using VGGNet convolutional model. Then the obtained features are filtered using a significantly enhanced Salp Swarm Algorithm (SESSA) is used to extract just useful characteristics and remove those that aren't necessary. We examined the proposed method on two different WBC Leukemia reference datasets and discovered it to be highly effective as well as being computationally simple. The outcomes are one of the best on these datasets, and they outshine a variety of CNN models. The proposed hybrid methodology succeeded magnificently in terms of accuracy and computational cost, which has a great impact on processing time and resource utilization.

In [2] S. Shafique, S. Tehsin, S. Anas and F. Masud proposed the concept, implementation, and analysis of a computerized screening system for AML in blood microscopic images are proposed in this paper. It makes use of 90 high-resolution 194X148 photos from the American Society of Hematology. The proposed system automates computational tasks such as nucleated cell segmentation, color correlation and accurate verification and segmentation. To extract all of the knowledge essential for

effective segmentation, a featured set incorporating the color, shape and texture parameters of a cell is constructed. The effect of the LBP operator on the HD was a promising feature for this research. In particular, a color component called cell energy was incorporated, and the results reveal that this feature effectively distinguishes cancer cells from non-cancer cells. Complete blood smear images are classified differently than subsets of images.

In [3] the suggested paper proposed by Acharya, Kumar, P. Tamim will investigate multiple computer-aided system approaches for categorizing microscopic blood smear images. The above-mentioned objective can be achieved in two ways. To commence, a unique algorithm for categorizing the nucleus and cytoplasm of white blood cells is established. Then, to extract the characteristics and train the model, a model is developed. In total, 600 photos were used in the experiment. The result demonstrates how acute lymphoblastic leukaemia is classified into three different categories: L1, L2 and L3. The model can tell the difference between a healthy and abnormal blood cell.

In [4], the purpose of this paper is to look into how to build image segmentation algorithms for acute leukaemia. White Cells in Blood Microscope Images are categorized and analyzed using strong segmentation and measurement methods. They used the Herlev Dataset and acquired a 94% accuracy. For feature extraction, a new method based on the GLCM methodology is utilized. To classify data and improve image processing, feature extraction is used. Prior to the development of statistical measurements, fractals have been used in a wide range of scientific domains to evaluate the properties of space prior to the emergence of statistical measurements. The histogram color approach is simple and clear to use, compute, and comprehend. The approach of the correlogram is spatial information.

In [7], a new methodology for the detection of cancer cells present in the human body, image processing techniques have been used. It was observed that Gradient Boosting Decision Tree classification algorithms gave better results than SVM. The pre-processing technique is used to identify and crop the ROI. In the next step, feature extraction is done and in order to get greater results from the processed images, various were used. In the classification model building step, 64 features have been extracted, which are analyzed further to check the relation between them. Depending upon the feature analysis, 45 features were selected to train the classifiers

III. PROPOSED SYSTEM

A Proposed System Architecture

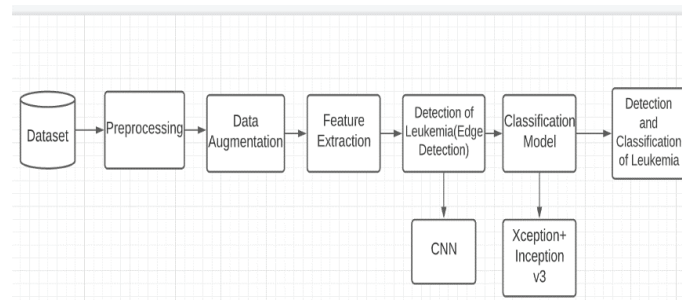


Fig. 1. Architecture diagram of the proposed system.

The above represents the proposed system architecture of leukemic detection and classification using transfer learning.

The system is responsible to take the blood smear image as input and give the type of leukemia to the user as output in case the presence of leukemia is detected.

A. Dataset Collection

Consider the ALL_IDB dataset from Kaggle which includes acute lymphoblastic images. It consists of 260 acute lymphoblastic and healthy images. The remaining types of leukemic images such as CML, CLL, AML are obtained from the ASH Imagebank. Image pre-processing is done on the images to make the images suitable for training. ASH Imagebank consists of very limited number of CML, CLL, AML microscopic blood smear images. So, Data Augmentation is performed on those images to increase the size of the dataset and to avoid overfitting of ALL type of data. The four categories of leukemia are shown below.

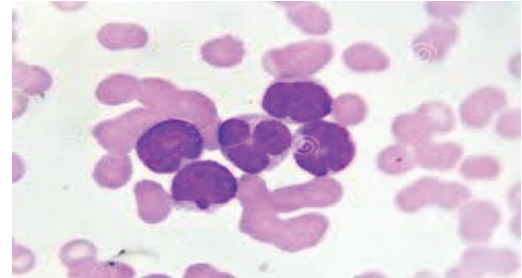


Fig. 2. Acute Myeloid Leukemia image

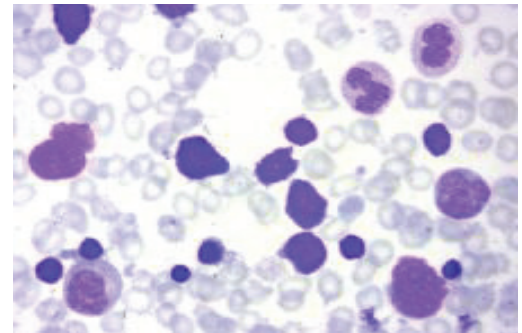


Fig. 3. Acute Lymphoblastic Leukemia

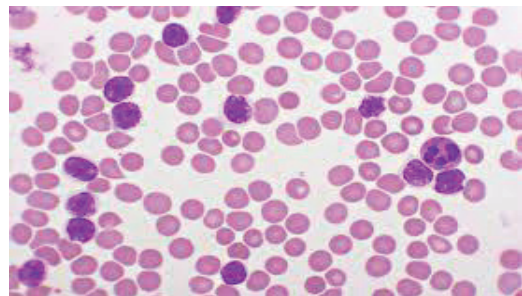


Fig. 4. Chronic Lymphocytic Leukemia

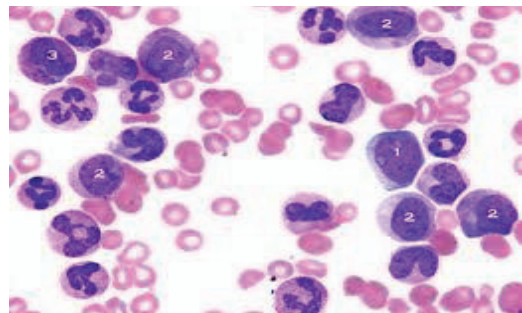


Fig. 5. Chronic Myeloid Leukemia

B. Leukemia Detection

Leukemic detection is performed by using image processing techniques. During the pre-processing of leukemic and healthy microscopic blood smear images, the images are reshaped into 224 X 224 size and then converted into gray scale format from the RGB format. Then various filters are applied on those images such as contrast stretching, Otsu thresholding to increase the contrast. Then edge detection segmentation is done on those images. For any image, edges are discontinuous local features. These features are used to locate the edges of leukemic cells with the help of filters and convolutions. One such well known filter to detect edges is Sobel filter which has two weight matrices. One weight matrix is used for detecting horizontal edges and the other one for detecting vertical edges. From the segmented image various features are extracted. The convolutional model trained with those leukemic and healthy images is used to classify the image as leukemic or healthy image and detects the boundaries of leukemic cells if leukemia is present.

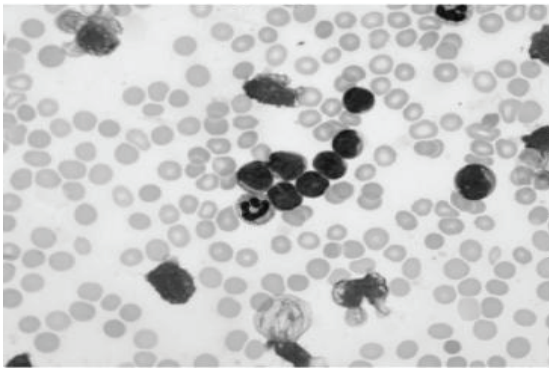


Fig. 6. Gray Image

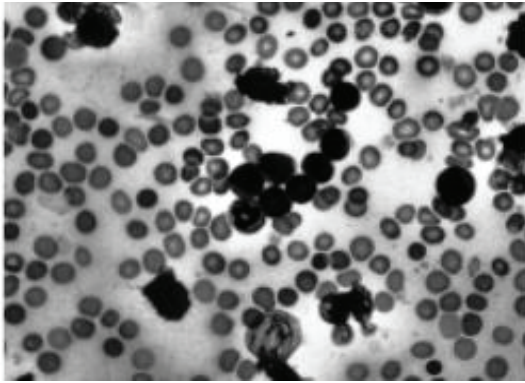


Fig. 7. Pre-processed image with increased contrast and gaussian blur filter

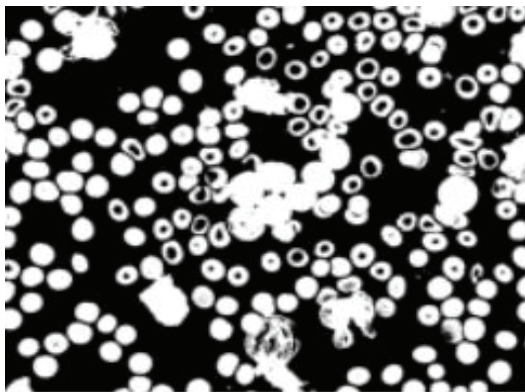


Fig. 8. Applied Otsu thresholding

C. Leukemic Classification into sub types

If leukemia is detected in the input image then it is classified into its sub types i.e acute lymphoblastic leukemia, acute myeloid leukemia, chronic myeloid leukemia, chronic lymphoblastic leukemia. A hybrid model is constructed in order to classify those leukemic images with better accuracy. The hybrid model is based on ensemble averaging technique. Ensemble averaging is a technique in which we can combine two or more models to improve the accuracy of output. It ensembles the Inception V3 and Xception convolutional models. The Inception V3 and Xception models are trained on the Imagenet and the knowledge obtained is transferred to the model constructed. The main idea behind using this transfer learning and ensemble averaging technique is that there is no need to construct the transfer learning models from scratch. The model obtained by training one kind of data can be used to perform classification on similar kinds of data. If the presence of leukemia is identified then this model developed using ensemble averaging will take the classification results from both Inception and Xception models and gives the output of the model which shows better accuracy. With this ensembled model the classification is performed. The GUI is developed by using Django which is a python web development framework. It will allow the user to enter their blood smear images and the developed model will tell what type of leukemia he has or healthy if no leukemia is detected in his blood sample.

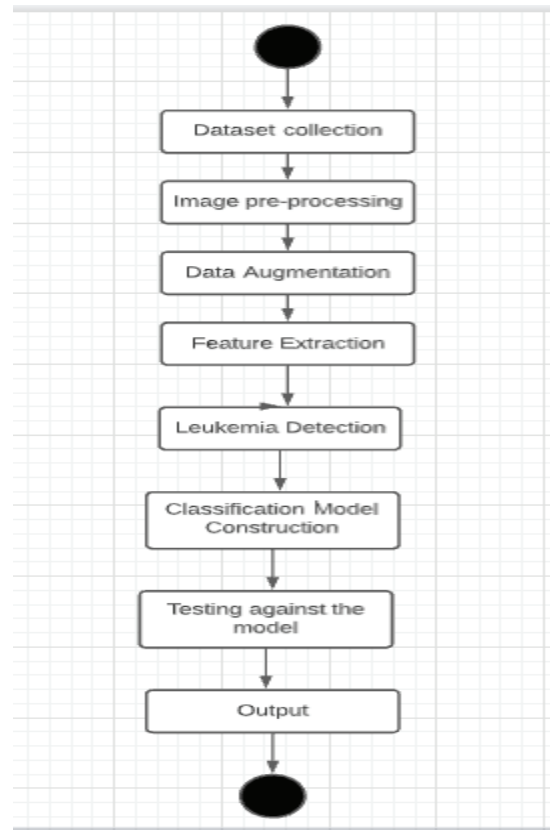


Fig. 9. Flow Chart of the proposed model

Fig 9 shows the flow chart of the proposed model. The flowchart starts with the input of blood smear image. Image preprocessing is performed and leukemia is detected by using canny edge detection. Then the leukemia is classified with the constructed classification

model. This is fed to GUI which gives the results to the user.

D. Algorithm

Step 1: Import all the required libraries.

Step 2: Convert the images present in RGB format to the gray scale images.

Step 3: Pre-process the images in order to convert them into more meaningful representation by applying Otsu thresholding and increasing contrast.

Step 4: Segmentation is performed on those images. Various features are extracted from those images and detection of leukemia is done with the help of convolutional neural networks

Step 5: A classification model is constructed based on transfer learning technique which uses Inception, Xception convolutional models.

Step 6: The inception V3 and Xception models are ensemble to form a hybrid model.

Step 6: The model is trained with different categories of leukemia.

Step 7: The ensemble averaging model will take the classification results from both Inception V3 and Xception convolutional models and gives the output of the model that has better accuracy.

Step 8: The graphical user interface developed using python will take the microscopic blood smear images and produces the classification results.

IV. IMPLEMENTATION

This section shows the outputs and results of the proposed system. A web application was developed using python django app which takes the blood smear image as input and classifies to what category of leukemia the image belongs to. The user uploads his blood smear image and once he clicks the submit button, he will get the output as healthy if he does not have leukemia or else what type of leukemia he has will be displayed. The input is an image and output is also an image showing the presence of leukemia with text indicating the type of leukemia.

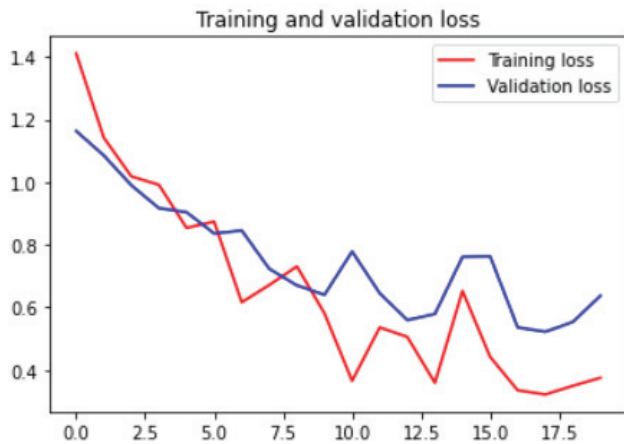


Fig. 10. Training and Validation loss using Xception

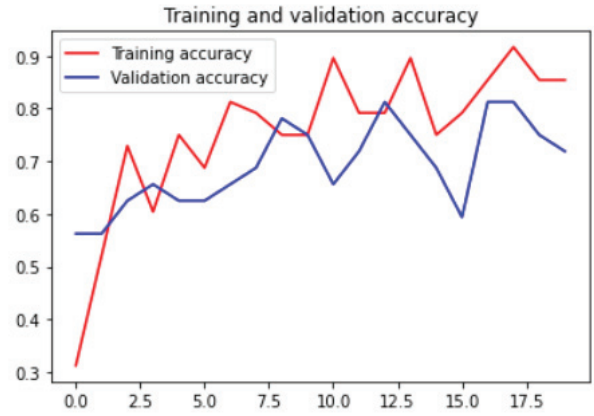


Fig. 11. Training and Validation accuracy using Xception

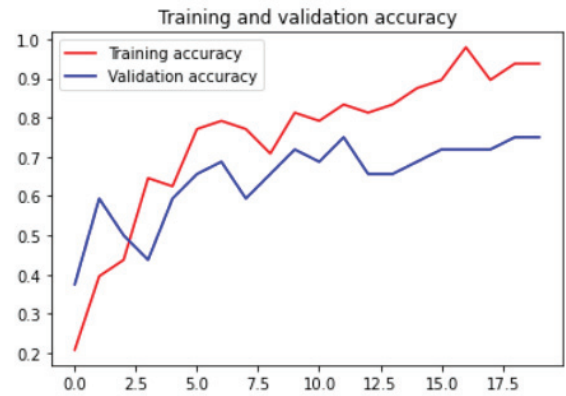


Fig. 12. Training and validation accuracy with Inception V3

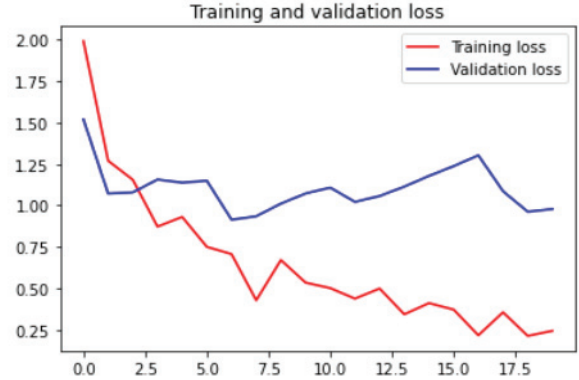
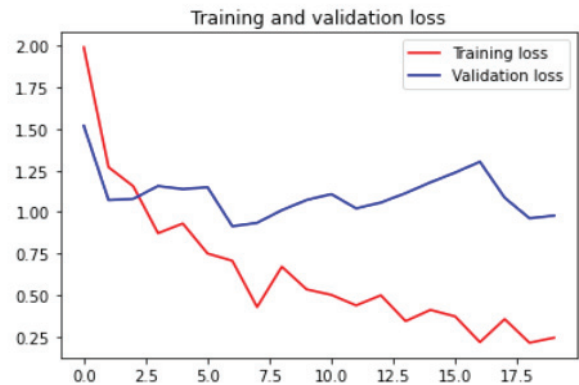


Fig. 13. Training and validation loss with Inception V3



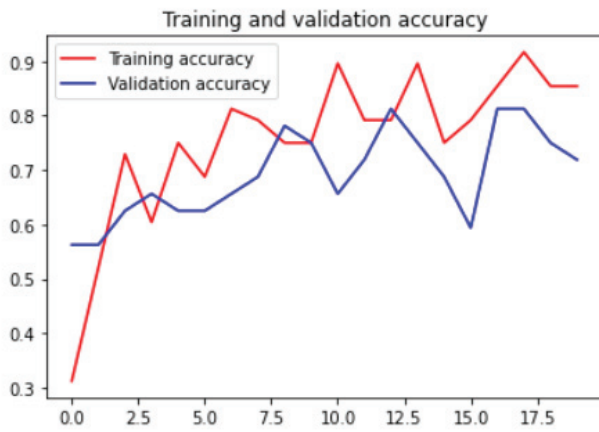


Fig. 14. represents the Training and validation loss, validation accuracy associated with the hybrid model constructed by ensembling Inception V3 and Xception convolutional models.

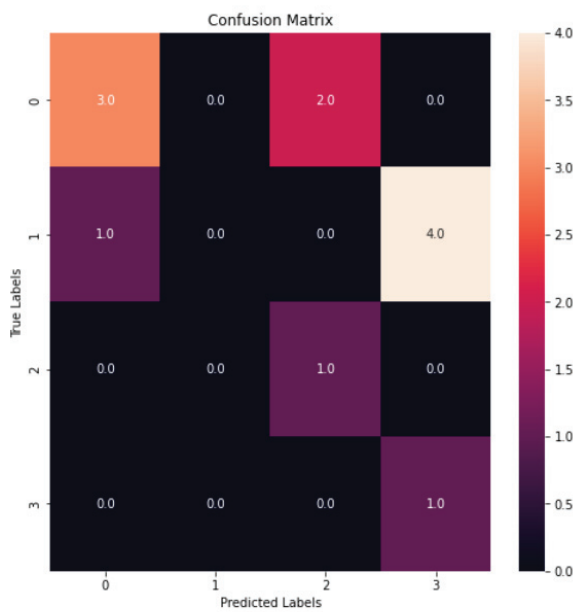


Fig. 15. Confusion matrix of the constructed hybrid model

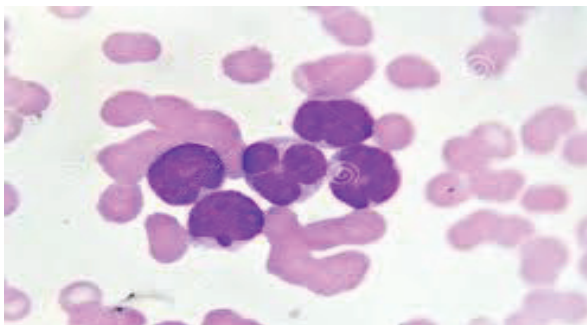


Fig. 16. Input AML image

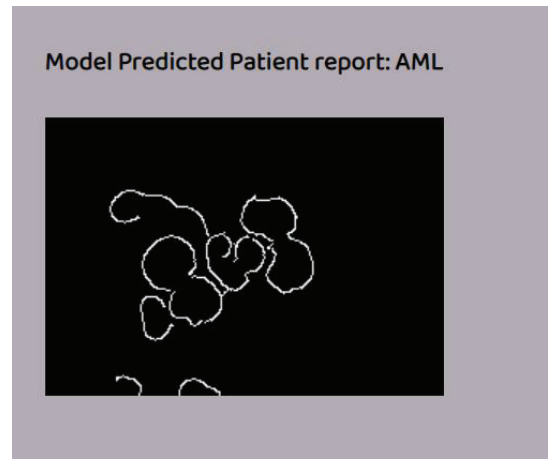


Fig. 17. GUI displaying the classification result

Fig 17 shows the presence of leukemia and what type of leukemia it is. Our model outperforms the other previously developed image processing models since this model uses ensemble averaging which is getting popular in terms of accuracy regarding classification related problems. Thus, the developed model using weighted averaging ensemble model achieved an accuracy of 91.71% in classifying the images which is very efficient when compared with previous models.

V. CONCLUSION AND FUTUREWORK

This project is very useful in early detection of leukemia. This project covers the identification of presence of leukemia and classification of it. This presently can detect whether a person has leukemia or not based on blood smear images. The result is displayed in a web page to make user friendly interface. This project implements both transfer learning and image processing techniques and gives the model which has best accuracy. It can be extended by Adding the module that can further classify and tell the sub types of leukemia if leukemia is detected. Accuracy of the model can also be increased by training with a large number of images. We can also improve our interface by adding more features.

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